ShopSphere Scalability Audit – Engineering Report

CSA06 – Design and Analysis of Algorithms | CO1 Assessment

Purpose

Assess whether ShopSphere's modules can scale from 100,000 to 1,000,000 products under a 10-minute nightly processing window, using the assignment's assumption of 10^8 basic operations per second.

Summary of findings

Module	Growth / illustrative estimate	Decision
M1 duplicate detector	$\Theta(n^2)$; ~83.33 min	Redesign
M2 rate limiter	$\Theta(n)$; 1,048,575 audits	Likely keep
S2 binary search	$\Theta(\log n)$; ~21 comparisons	Keep sorted index
M4 current	$\Theta(n^2)$; ~333 min estimate	Redesign
M4 redesign	$\Theta(n \log n)$; ~0.21 s estimate	Prefer if valid
M5 recurrence	$\Theta(n \log^2 n)$; ~3.97 s scale estimate	Benchmark
Report f(n)	$\Theta(n^2)$; ~500 min estimate	Redesign

Recommendations

1. Replace M1's all-pairs scan with hash-based grouping on seller and normalized title. This changes expected work from $\Theta(n^2)$ to $\Theta(n)$, at the cost of additional memory.

2. Consider M4's non-replicated design. Its growth improves from $\Theta(n^2)$ to $\Theta(n \log n)$, but validate that removing replication preserves the required results.

3. Redesign the nightly report generator: the leading $3n^2$ term dominates and exceeds the 10-minute window. Consider incremental aggregation, indexing, or processing only changed sales records.

4. M2 and S2 appear scalable by operation count, but validate actual throughput, I/O, concurrency, and implementation overhead.

Evidence note

The CSV and output files contain calculated formula-based estimates, not production measurements. The assignment's exact M2 count at n=1,024 is 1,023, and M1's exact count is $n(n–1)/2$. Measure actual timings before making deployment decisions.